# ENVIRONMENTAL PERFORMANCE, LCA AND ESG SUSTAINABILITY REPORT

Project: **Example - tutorial house in London (ERA5 1991-2020) [costs to be entered by the user]**

Life Cycle Assessment (LCA), Carbon Footprint and Sustainable Certifications

⚠ **The grid emission factor was not informed: the emissions from pumping electricity are not meaningful until it is entered.**

⚠ **Investment, maintenance and electricity costs were not informed: the economic result shows only the savings on the water bill; payback and NPV are not meaningful until costs are entered.**

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Triton 0.1.0 · International Edition (INT)

**ISO 14040 / ISO 14044 · GHG Protocol**

| CLIMATE LOCATION | CLIMATE BASIS | SERIES PERIOD | TARIFF LOCATION |
|---|---|---|---|
| London / United Kingdom | Observed series | 01/01/1991 – 31/12/2020 | - |

| TANK / SAVINGS | EMBODIED CARBON | AVOIDED CO2 / YR | 20-YEAR NET CARBON | CARBON PAYBACK |
|---|---|---|---|---|
| **1,000 L** | **160.4 kg** | **+0.1 kg/yr** | **+163 kg** | **> 50 years** |
| **14.5%** potable replaced | Materials & Equipment | Net Annual Balance | Lifetime Net Mitigation | -0.26 g CO2e/L avoided |

## 1. 1. BUILDING & DEMAND INPUTS

| Parameter | Value |
|---|---|
| Rainfall Series | **London.csv** |
| Catchment Area (Roof) | **150.0 m²** |
| Runoff Coefficient (C) | 0.85 |
| First Flush Discard | 2.0 mm |
| Occupants | 4 occupants |
| Per-capita Demand | 150 L/person/day |
| Rainwater Target Share | **28%** |

## 2. 2. HYDRAULIC BALANCE (YAS MODEL)

| Hydraulic Indicator | Result |
|---|---|
| Storage Tank Volume | **1,000 litres** |
| Potable Savings | **14.5%** |
| Annual Pluvial Use | **31.7 m³/yr (86.8 L/day)** |
| Residual Potable Demand | 187.5 m³/yr (513.2 L/day) |
| Annual Overflow | 17.8 m³/yr (48.7 L/day) |
| Full Reliability Days | **46.3% of days** |
| Total Water Demand | 219.1 m³/yr |

## 3. 3. LIFE CYCLE INVENTORY (LCI — ISO 14040/14044)

| Component / Phase | Quantity / Size | Embodied CO2 | Op. CO2/Yr | 20-Yr Balance | Source / Reference |
|---|---|---|---|---|---|
| **Lower Cistern (Polyethylene)** | 1,000 L (22.0 kg) | **+35.5 kg** | -- | **+35.5 kg** | Resin: Franklin Associates for the American Chemistry Council (out/2020), 'Cradle-to-Gate Life Cycle Analysis of High-Density Polyethylene (HDPE) Resin', Table 6: 1,612 kg CO2e per 1,000 kg of resin (TRACI 2.1 / IPCC 2013 GWP-100, US scope). Replaces the previous internal value of 3.00, which had no source and declared the same boundary. The mass per $m^3$ remains an internal scenario. |
| **Rigid PVC Pipe (DN 32 mm)** | 20.0 m (0.71 kg CO2/m) | **+14.1 kg** | -- | **+14.1 kg** | Manufacturer's data sheet (2025), solvent-weld pipe 32 x 6 m: 1,793.14 g × 2.36 kg CO2e/kg (EPD S-P-05503, RXP PVC Pipes, A1-A3) |
| **Rigid PVC Pipe (DN 40 mm)** | 5.0 m (1.01 kg CO2/m) | **+5.0 kg** | -- | **+5.0 kg** | Manufacturer's data sheet (2025), solvent-weld pipe 40 x 6 m: 2,561.27 g × 2.36 kg CO2e/kg (EPD S-P-05503, RXP PVC Pipes, A1-A3) |
| **Rigid PVC Pipe (DN 100 mm)** | 20.0 m (2.78 kg CO2/m) | **+55.5 kg** | -- | **+55.5 kg** | Standardised pipe geometry, reinforced series: OD 101.6 mm, wall 2.5 mm × 2.36 kg CO2e/kg (EPD S-P-05503, RXP PVC Pipes, A1-A3) |
| **Pressurising Pump (0.24 CV - Cast Iron, Copper & Aluminium)** | 1 pcs (main) — 0.24 CV | **+29.1 kg** | -- | **+29.1 kg** | Manufacturer catalogue mass (60 Hz) × 4.77 kg CO2e/kg from the Grundfos EPD EPD-GRU-20250340-CBA1-EN (A1-A3): order-of-magnitude approximation; the EPD of the manufacturer actually used prevails |
| **First-Flush Discard System (Leaf Filter + Tank 300 L)** | 300 L (9.3 kg material) | **+21.1 kg** | -- | **+21.1 kg** | First-flush discard / Polymer & Stainless-steel inventory |
| **Pumping Electricity Use** | 29.3 kWh/yr (0.93 kWh/$m^3$) | **--** | -- | -- | Editable grid factor; use the factor of the site and of the study period |
| **Treatment Line** | 31.7 $m^3$/yr; 0,073 kg product/yr (1.50 mg Cl2/L; 0,004 kg CO2e/$m^3$) | **--** | **+0.1 kg/yr** | **+2.5 kg** | Stages: Chlorination (calcium hypochlorite tablet feeder). Sized by the Triton water-quality module |
| **Mains Potable Water Avoided** | 31.7 $m^3$/yr (0.00 kg CO2/$m^3$) | **--** | -- | -- | Editable potable-water factor; replace with the inventory of the entity that supplies the site |

## 4. 4. CARBON BALANCE, ECO-PAYBACK & METHODOLOGY

| Total Embodied Carbon | Annual Pumping | Chlorination / Treatment | Avoided Water Credit | Net Annual Saved | Eco-Payback |
|---|---|---|---|---|---|
| **+160.4 kg CO2e** | **+0.0 kg/yr** | **+0.1 kg/yr (0,073 kg product/yr at 1.50 mg Cl2/L)** | **-0.0 kg/yr** | **+0.1 kg CO2e/yr** | **> 50 years** |

**Carbon Balance Formulation & Convention:** *Official convention: Carbon emitted to atmosphere is POSITIVE (+) in red (environmental impact); carbon avoided/removed from atmosphere is NEGATIVE (-) in green (ecological benefit).*
*20-Year Net Balance (kg CO2e) = Embodied CO2 (+) + [Pumping CO2 (+) + Treatment CO2 (+) - Avoided Grid Water (-)] × 20 yrs.* Environmental payback represents the time required for operational avoided emissions to offset initial material embodied carbon.

**Treatment train considered:** Chlorination (calcium hypochlorite tablet feeder).

## 5. 5. ESG RATING & GREEN BUILDING CERTIFICATIONS

### 🏛️ LEED v4 / v4.1 (WE)

**0 of 8 points (Max: 8 pts)**

**Scored 0/8 pts (Below the LEED threshold (min. 25%))**

WEc1 (Ext.): 0/2 pts | WEc2 (Int.): 0/6 pts

### 🏛️ LEED v5 BD+C (WEc2)

**2 of 8 points**

**Options 2 to 6**

Option 2: 0 | 4: 0 | 6: 2 | 1: 0

### 🌎 EDGE (IFC)

**Water bar: 14.5%**

**Short by 5.5 p.p.**

No points: EDGE measures percentage against a local base case. WEM14 catchment area: 150 $m^2$

### 💧 GRI 303 & Water Positive

**31.7 $m^3$/yr** preserved

**Ratio: 0.14x**

633 $m^3$ over 20 yrs

### 🌿 ESG Executive Synthesis & Decarbonization

The project reaches an overall potable-water substitution of 14.5% (indoor: 14.5%, outdoor: 0.0%), saving 32 $m^3$/yr of treated water (633 $m^3$ accumulated over 20 years). Under LEED v4 WE (maximum of 8 points), the project scored 0 of 8 points. For EDGE, which awards no points but measures a percentage against a baseline case, the water bar reaches 14.5% — below the 20% threshold required in each of the three bars. The rainwater harvesting system plays a vital role in urban water security and climate resilience, contributing directly to Scope 3 GHG mitigation.

### ⚠️ Technical Responsibility & Methodological Notice:

This environmental report was generated by Triton following the applicable technical standards and the ISO 14040/14044 guidelines. Emission factors represent screening engineering scenarios and must be validated by the responsible environmental professional against supplier EPDs and the parameters of the local grid and water supply.

Rainfall: weather data from Open-Meteo.com (CC BY 4.0), based on the ERA5 reanalysis of the Copernicus Climate Change Service (Hersbach et al., 2020).